

DIEGO MAGANA

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Principal Engineer with ~10 years across data science, ML, and engineering, now pursuing an MSCS (AI · GPA 3.93). Builds deep-learning systems from first principles — autograd, language models, transformers — and investigates their internal mechanisms through causal interpretability. Focused on mechanistic interpretability, alignment, and building systems whose behavior can be explained, not just measured.

EDUCATION

Johns Hopkins University

Baltimore, MD

Master of Science, Computer Science (AI) — in progress · GPA 3.93

Amherst College

Amherst, MA

Bachelor of Arts (2017) · NCAA Division III Men's Basketball — NESCAC Champion, NCAA Semifinalist (2014)

ML PROJECTS

gpt — Mechanistic Interpretability of a Transformer · github.com/diego-magana/gpt

- Found that a single attention head carries a character-level transformer's next token prediction and isolated it with convergent causal evidence — head ablation (+0.89 nats, 7× the next head) and single-head activation patching (~0.51 recovery, 5× the next) — on cache, ablation, and patching instrumentation built from scratch.
- Confirmed the mechanism with a pairwise-ablation redundancy test and replicated it across five training seeds, finding it robust but its head index an initialization artifact; tested the boundary with a corruption-distance sweep and reported the architecture's limits explicitly rather than over-claiming generality.

micrograd — Autograd Engine from First Principles · github.com/diego-magana/micrograd

- Built a scalar-valued reverse-mode autodiff engine from scratch — dynamic graph via operator overloading — with gradients verified against PyTorch to 1e-5 across all operators; trained an MLP (337 params) to 100% accuracy on `make_moons`.

makemore — Character-Level Language Model Series · github.com/diego-magana/makemore

- Implemented the full character-level progression — bigram → MLP (Bengio 2003) → WaveNet (77K params, 2.01 validation NLL) — including a from-scratch backprop variant matching PyTorch autograd to within ~0.001 nats.
- Authored an original probing analysis of the trained model's representations — embedding geometry, gradient-based context sensitivity, and single-unit ablation — with explicit reporting of correlational vs. causal limits.

EXPERIENCE

Raxwell Inc., Irvine, CA

Jan '23 – Present

Principal Engineer (Contract)

Oct '25 – Present

- Built a component-separated RAG evaluation harness benchmarking open models (Llama 3.1, Mistral) against managed copilots on Business Central + M365 workflows — scoring retrieval, faithfulness, and numerical accuracy as separate gates — producing the build-vs-buy cost model.
- Instrumented the pipeline end-to-end (per-component tracing, CI regression gate), diagnosing failures by isolating retrieval miss from generation drift.
- Sole technical owner of the company's Microsoft and Business Central stack — architecture, deployment, and production operations.

Chief Information Officer

Mar '24 – Oct '25

- Owned end-to-end technology strategy and execution through a growth phase — Azure cloud, enterprise data infrastructure, and the M365 ecosystem — as the senior-most technical authority in the organization.
- Delivered enterprise system integrations across finance, supply chain, and EDI — connecting Business Central with external trading-partner and operational systems as the backbone of daily operations.

Principal Engineer (Contract)

Jan '23 – Mar '24

- Architected the company's Azure migration and built the data platform and ML/analytics infrastructure still in production today.
- Built ETL pipelines, REST API integrations, and Power BI dashboards adopted as primary operational decision tooling.

Mavco LLC, Scottsdale, AZ

Aug '17 – Jan '23

Senior Data Scientist

Aug '20 – Jan '23

- Established the company's first analytics and ML function, scaling the modeled catalog from hundreds of SKUs to tens of thousands via automated pipelines.
- Owned the full ML lifecycle for forecasting, pricing, and classification systems — data engineering through production deployment — driving a ~10-point lift in adjusted gross margin and informing seven-figure capital allocation.

Data Scientist

Aug '17 – Aug '20

- Built ML pipelines for sales and pricing prediction and developed financial models informing executive strategy.

TECHNICAL SKILLS

- **Deep Learning:** PyTorch, transformer architectures, LLM pretraining & fine-tuning, language modeling
- **Interpretability & Analysis:** activation/residual/head patching, logit lens, head ablation; causal vs. correlational evaluation
- **ML Systems & Evaluation:** RAG evaluation harnesses, component-separated metrics, build-vs-buy cost modeling
- **Machine Learning:** scikit-learn, time-series forecasting, classification, statistical modeling
- **Infrastructure & MLOps:** Docker, GitHub Actions (CI/CD), Modal, W&B, Azure AI Foundry
- **Languages:** Python, Java, SQL, C++